

Introduction to Programming Paradigms

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1 Programming Paradigms

The logical approach in software engineering is descriptive of how programming language is implemented.

1.1 Reasons for studying programming paradigms

- Increased ability to express ideas
- Improved background for choose appropriate language
- Increased ability for learn new languages

2 Programming Domains

- Scientific Computing : Fortran
- Business Applications : COBOL
- AI : LISP
- Systems Programming : C/C++
- Web Software : Markup languages (HTML/CSS) , Scripting languages (JavaScript , PHP) or General-Purpose languages (Java , Python)

3 Language Evalutation Criteria

- **Readability** : Programs can read and understood
- **Writability** : Language can be used to create programs
- **Reliability** : Conformance to specifications
- **Cost** : The ultimate total cost

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3.1 Evaluation Criteria: Readability

- **Overall Simplicity** : manageable set of features and constructs
- **Orthogonality**
- **Data types** : Adequate predefined data types
- **Syntax Considerations**

3.2 Evaluation Criteria: Writability

- **Simplicity and Orthogonality**
- **Support of Abstraction**

3.3 Evaluation Criteria: Reliability

- **Type checking**
- **Exception Handling**
- **Aliasing**
- **Readability and Writability**

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